

Regular Expression Practice Questions

1. Write a regular expression to extract all email addresses from a given text. Assume the email format follows the standard convention: username@domain.extension (e.g., john.doe@example.com).
2. Given a tokenized sentence, write a regular expression to filter out all tokens that are purely numeric (e.g., 2023, 123), while retaining tokens that contain both letters and numbers (e.g., abc123).
3. Create a regular expression to detect repeated consecutive words in a sentence, such as "the the" or "you you". Ensure the match is case-insensitive.
4. Design a regular expression to extract the first capitalized word from each English sentence. You may assume that each sentence ends with a period (.), question mark (?), or exclamation mark (!).
5. Write a regular expression to identify and extract all dates in the format dd-mm-yyyy or dd/mm/yyyy from a given document.
6. Write a regex to extract all valid Indian phone numbers from a given text. Ensure your regex can handle formats like +91-9876543210, 09876543210, or 9876543210, including optional spaces or dashes.
7. Write a regex to validate Indian vehicle registration numbers (e.g., MH12AB1234).
8. Write a regex to validate an Indian Aadhar number in the format XXXX XXXX XXXX (e.g., 1234 5678 9123).
9. Write a regex to validate Indian PAN card numbers (e.g., ABCDE1234F).
10. Write a regex to extract Indian date formats such as DD/MM/YYYY, DD-MM-YYYY, and D/M/YY, including mixed delimiters and optional leading zeros.
11. Write a regex to extract Indian postal codes (6-digit numbers like 110001).
12. Write a regex to extract Indian currency formats like ₹1,00,000, Rs. 100000, or INR 1,00,000.
13. Write a regex to extract valid time formats including 12-hour (e.g., 02:30 PM) and 24-hour (e.g., 14:30) styles.
14. Write a regex to validate Indian GST numbers in the format 22AAAAA0000A1Z5.
15. Write a regex to validate and extract Indian IFSC codes in the format XXXX0YYYYYY.
16. Write a regex to extract dates written in words (e.g., 1st January 2022, 25th December).
17. Write a regex to match valid hexadecimal color codes (e.g., #FFF, #ffffff).

18. Write a regex to extract all valid IPv4 addresses from a text.
19. Write a regex to extract hashtags from social media text (e.g., #AI, #NLP2025).
20. Write a regex to extract mentions in social media text (e.g., @username).
21. Write a regex to identify strings that are valid Python variable names.
22. Write a regex to extract valid credit card numbers (e.g., 1234-5678-9012-3456).
23. Write a regex to extract all HTML tags from a document.
24. Write a regex to extract domain names from URLs (e.g., www.example.com).
25. Write a regex to identify lines that begin with a digit.
26. Write a regex to extract MAC addresses (e.g., 00:1A:2B:3C:4D:5E).
27. Write a regex to extract file names with extensions .pdf, .docx, or .txt.
28. Write a regex to validate strong passwords (at least one uppercase, one lowercase, one digit, one symbol, and 8+ characters).
29. Write a regex to extract temperature values in Celsius or Fahrenheit (e.g., 32°C, 100F).
30. Write a regex to match binary numbers that are divisible by 4.
31. Write a regex to match valid Roman numerals.
32. Write a regex to find words longer than 10 characters.
33. Write a regex to extract phone numbers with country code and parentheses (e.g., +91 (98765) 43210).
34. Write a regex to extract numerical ranges like 1-10, 100-200 from a string.